

Barbalat's Lemma

Arun Verma

20 June 2026

A Note from the Author:

Hey everyone!

I was messing around with that classic problem where $\lim_{x \rightarrow \infty} (f'(x) + f(x)) = 0$ (the one listed as Problem 1 below).

Usually, we just mindlessly throw the e^x integrating factor trick at it, but I started thinking about the actual geometry behind it. Trying to visualize why the function can't just oscillate forever or throw up weird, sharp spikes at infinity led me down a total rabbit hole—and I stumbled across this super cool real analysis tool called **Barbalat's Lemma**!

It completely flipped how I think about uniform continuity. It's not just some dry property we use for compact sets; it basically acts like a global speed limit that tames how wildly a function can fluctuate at infinity.

I got super inspired and cooked up a few original problems based on this exact machinery. They blend things we already know—like the Heine-Cantor theorem and shifting sequences—but in ways that really make you stretch your intuition about uniform continuity and asymptotic decay.

I put them together in this file. Definitely try the first problem to see the spark that started it all! Let me know if you manage to crack them or which one breaks your brain first.

Theorem 1 (Barbalat's Lemma). *Let $f : [0, \infty) \rightarrow \mathbf{R}$ be a differentiable function.*

if:

1. *$f(t)$ has finite limit as $t \rightarrow \infty$ (i.e., $\lim_{t \rightarrow \infty} f(t) = c < \infty$)*

2. *The derivative $f'(t)$ is uniformly continuous*

Then:

$$\lim_{t \rightarrow \infty} f'(t) = 0$$

Proof. .

□

Ques 1. *Let $f : (0, \infty) \rightarrow (0, \infty)$ be a continuously differentiable function: this means that f is differentiable at x for all $x \in (0, \infty)$, and that $f' : (0, \infty) \rightarrow \mathbf{R}$ is continuous. Assume that*

$$\lim_{x \rightarrow \infty} (f'(x) + f(x)) = 0$$

Show that $\lim_{x \rightarrow \infty} f(x) = 0$

(Alternative solution)

consider, for any $x \in (0, \infty)$ we can write, $f(x) = \frac{e^x f(x)}{e^x}$, now let $g(x) = e^x f(x)$ and $h(x) = e^x$, we want to evaluate the limit, $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{g(x)}{h(x)}$

Theorem 2 ((L'Hopital's Rule at infinity)). : Let $g(x)$ and $h(x)$ be differentiable functions on an open interval (a, ∞) where $h'(x) \neq 0$. if $\lim_{x \rightarrow \infty} h(x) = \infty$ (or $-\infty$) and the limit of their derivatives exists:

$$\lim_{x \rightarrow \infty} \frac{g'(x)}{h'(x)} = L \quad (\text{where } L \in \mathbf{R}).$$

Then the limit of $\frac{g(x)}{h(x)}$ exists and equal to L i.e $\lim_{x \rightarrow \infty} \frac{g(x)}{h(x)} = L$

(Reference: Rudin Chapter 5 Theorem 5.13)

Now, evaluate $\lim_{x \rightarrow \infty} \frac{g'(x)}{h'(x)} = \lim_{x \rightarrow \infty} \frac{e^x f'(x) + e^x f(x)}{e^x} = \lim_{x \rightarrow \infty} \frac{e^x (f'(x) + f(x))}{e^x} = \lim_{x \rightarrow \infty} (f'(x) + f(x)) = 0$

By L'Hopital rule the limit of $\frac{g(x)}{h(x)}$ exists and equal to 0. Hence, $\lim_{x \rightarrow \infty} f(x) = 0$

Ques 2. Let $f : [0, \infty) \rightarrow \mathbf{R}$ be a twice-differentiable function. Suppose that: $\lim_{x \rightarrow \infty} f(x) = L$ (where L is a finite number). The second derivative $f''(x)$ is bounded for all $x \in [0, \infty)$ prove that $\lim_{x \rightarrow \infty} f'(x) = 0$

Ques 3. Let $f : (0, \infty) \rightarrow (0, \infty)$ be twice continuously differentiable function. Assume that: $\lim_{x \rightarrow \infty} (f''(x) + 2f'(x) + f(x)) = 0$. Show that $\lim_{x \rightarrow \infty} f(x) = 0$ and $\lim_{x \rightarrow \infty} f'(x) = 0$

Ques 4. Let $f : [0, \infty) \rightarrow \mathbf{R}$ be continuously differentiable function. Suppose that the improper integral $\int_0^\infty f(x)^2 dx$ converges, and its derivative is bounded ($|f'(x)|$, for all $x \geq 0$)

Prove that:

$$\lim_{x \rightarrow \infty} f(x) = 0$$

Ques 5. Let $f : [0, \infty) \rightarrow \mathbf{R}$ be a differentiable function such that $\lim_{x \rightarrow \infty} f(x) = c$ (where $c \neq 0$ is a finite constant). if the function $g(x) = f(x)f'(x)$ is uniformly continuous, prove that: $\lim_{x \rightarrow \infty} f'(x) = 0$

Ques 6. Let $f : [0, \infty) \rightarrow \mathbf{R}$ be a differentiable function such that $\lim_{x \rightarrow \infty} f(x) = 0$. Suppose there exist constant $k > 0$ and $0 < p < 1$ such that the derivative satisfies: $|f'(x)| \leq k|f(x)|^p \quad \forall x \geq 0$. prove that $\lim_{x \rightarrow \infty} f'(x) = 0$

Ques 7. Let $f : [0, \infty) \rightarrow \mathbf{R}$ be a twice-differentiable function such that $\lim_{x \rightarrow \infty} f(x) = 0$ and its second derivative is bounded ($|f''(x)|$, for all $x \geq 0$). For each $n \in \mathbf{N}$, define a sequence of functions on the interval $[0, 1]$ by:
 $g_n(x) = f'(x + n)$

1. prove that the seq $\{g_n\}$ is uniformly equicontinuous on $[0, 1]$
2. prove that the seq $\{g_n\}_{n=1}^{\infty}$ converges uniformly on $[0, 1]$ and find its limit function.

Ques 8. Suppose $f : [0, \infty) \rightarrow \mathbf{R}$ be a differentiable function and f' is uniformly continuous on $[0, \infty]$. Assume further that the following improper integral converges:

$$\int_0^{\infty} f(x)f'(x)e^{-x}dx < \infty$$

if you are given that $\lim_{x \rightarrow \infty} f(x)e^{-x} = 0$, prove that:

$$\lim_{x \rightarrow \infty} f(x)f'(x)e^{-x} = 0$$